VF Asia Footwear Laboratory Testing Manual

Test Method for Outsole Belt (Bata Belt) Flex	Test method # VF- D03 Page 1 of 5
Date Issued : October 13 , 2014 Revision # :	Created by : Nila Babia Erazo / Chris Adam

PURPOSE AND SCOPE: To measure the resistance of outsoles to repeated flexing to simulate the wear conditions of a shoe outsole

REFERENCE: SATRA TM 133

EQUIPMENT and MATERIALS:

1. Bata Belt abrasion machine, meeting the requirements of SATRA 133.
2. Belt of cotton canvas
3. Cutting devise, such as sharp knife or scalpel
4. T-square, with an integral angle of 90 ±1 °
5. Caliper to measure the crack
6. Buffing machine
7. Adhesives: A pre-reactive PU adhesive OR a single component PU adhesive.
8. Drying oven capable of heating a dry adhesive film
9. Platen press with the capability of applying a pressure of 500 ±50 kpa over the surface of the specimen
10. Outsole to be tested

SAMPLE PREPARATION:

Condition the sample for at least 48 hours at standard laboratory environment at 23.0±2.0˚C temperature and 55 ±10 % Relative Humidity.

1. If the specimen is an outsole attached to an upper or straps :
   1.1. Cut off the upper or straps just above the level of the insole
   1.2. Remove the insole
   1.3. Scour or flatten the lasted margin of the upper by roughing. Ensure not to damage the sole.

Notes:

- Do not use heat to soften the bond of the sole to the upper as this may damage the sole
- If the edge of the forepart on the reverse side is cupped or includes any form of rand or imitation welt, sour of buff this off until this surface is flat.
- For soles that are thicker than 15mm, reduce the thickness to 15mm using the buffing machine.

2. Make a center line from the toe to heel on the outer surface of the outsole as follow in Figure 1.

   2.1. Place the points A and B, see figure 1, on the inside edge of the outsole against the longer edge of the T square and the toe against the shorter edge.

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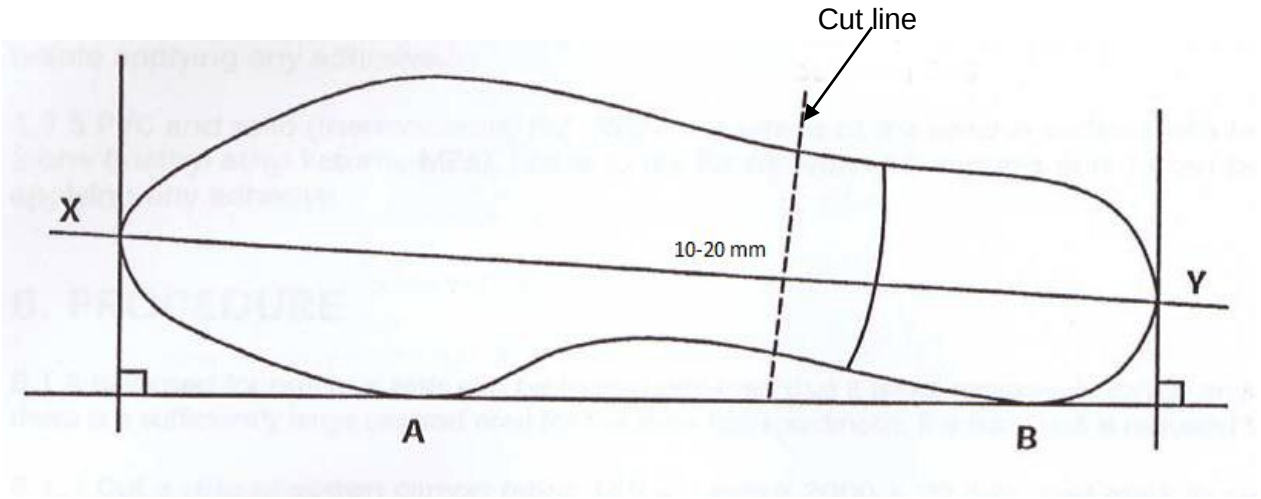
2.2. Make a mark on the toe at the point that is in contact with the shorter edge of the T square.

2.3. Repeat the procedure in the sections 2.1 and 2.2 for the heel of the specimen.

2.4. Mark a line between the point at the toe and the point at the heel, i.e. XY in figure 1.

2.5. Cut off the heel and part of the waist portion so as to leave 10 to 20mm length of the waist on the forepart as shown in figure 1.

Figure 1



3. A belt for previous tests can be used again provided that it is not excessively frayed and there is a sufficiently large unused area for the new specimens. If a new belt is required, then:

3.1. Cut a strip of canvas fabric of 140 ± 5 mm × 2000 ± 20 mm, and mark its center.

3.2. Draw two lines across the strip of the fabric:

3.2.1. At 90° to the longer edges of the strip.
3.2.2. One 930 ± 3mm on either side from the center of the strip.
3.2.3. One is on the upper surface and the other is on the lower surface.

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4. Place the specimens to be tested onto the belt, see figure 2:

4.1. Their wearing surfaces are uppermost.
4.2 They are evenly spaced between the two lines drawn in section 3.2.
4.3 They are no closer than 50mm to each other or the two lines drawn in section 3.2.
4.4. Their toes are facing to the larger roller.
4.5 Their toes are pointing in the same direction.
4.6 Their centerline is parallel to the longer edge of the belt.
4.7. They do not cover any previously used areas of the belt.

5. Mark the perimeter of each specimen on the belt.

6. Apply the polyurethane based adhesive to the surface of the belt over:

6.1. The whole area of each of the marked specimen positions.
6.2. The area between the two lines drawn in section 3.2 and the ends of the belt.

7. Apply the polyurethane based adhesive to the prepared surface of each specimen. Make sure that the specimens have been dried about 15minutes to 8 hours before applying the adhesive.

8. Use the drying oven to heat the adhesive coated specimens and belt .Temperature depends on the type of cement used, usually at 55°C to 80°C.

9. Attach the specimen immediately to the adhesive coated area of the belt so that the specimen is aligned with the perimeter drawn on the belt. Use a press to apply a pressure of 500±50kpa to bond the specimen to the belt.

10. Immediately place the two ends of the belt together so that the lines drawn in section 3.2 are aligned. Apply enough pressure by hand to bond the belt together. If necessary, can sew the two ends of the belt together.

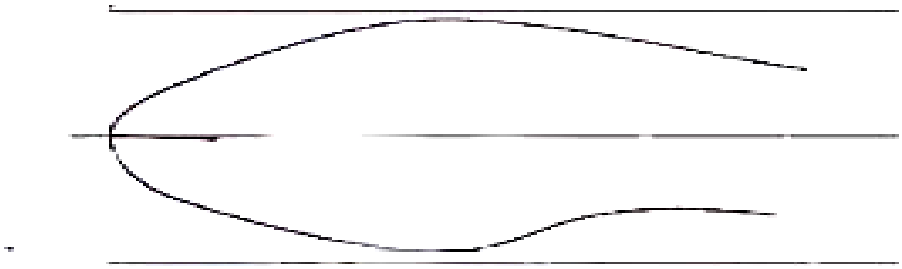
11. Condition or store the prepared belt in standard controlled environment of 23±2℃ and 55±10% relative humidity for more than 16 hours to allow the adhesive to set before your test.

Note:

If the specimens are particularly hard or stiff, it may also necessary to fasten the specimen and the belt together by a row of stitching.

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Figure 2



REPETITIONS: Test two (2) samples per material unless there's no available sample for the second test.

TEST PROCEDURE:

1. Select a proper smaller roller as per sole units.

    1.1. Loosen the setscrew (see figure1-3) and install a selected roller to the shaft.

Note:

- Select a suitable smaller roller before your test as follows:
- 90mm in diameter suitable for typical sole units.
- 60mm in diameter suitable for very flexible sole units.
- 120mm in diameter suitable for very rigid sole units.

2. Fit the prepared belt over the larger roller, and the smaller roller. When installing, can turn the crank to move the smaller roller so as to facilitate the installation of the belt. After that, turn the crank again to make the belt taut.

3. Connect a selected power supply to the machine according to the description on the nameplate.

4. Set the number of flexing cycles according to the test requirement

5. Press the START button and when the set value has been reached, the machine will stop automatically.

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Notes:

- During the test, check the bond between the ends of each of the specimen and the belt. If the length of the separation between the toe and the belt is more than 10mm, it is recommended that the test be interrupted while the affected area of the specimen is reattached.
- Before testing, close the acryl mask or the machine cover, otherwise the machine will not start.
- During the operation, do not open the acryl mask.
- If you want to inspect the specimens half way, press the STOP button.
- Inspect the specimens in every 5,000 cycles, until the required number of cycles is reached.

6. Remove the specimens from the belt. If it is difficult to remove try heating the adhesive bond from the belt side with the drying oven to a temperature between 80°C and 85°C for approximately 40 seconds.

   Note: The adhesive should be soft enough for the specimens to be peeled off.

7. Slowly move the belt round by hand and inspect the surface of each specimen for signs of cracking as it passes over the smaller roller. If any cracks are found, then record their position, and their length and depth to the nearest 1mm.

RERORTS:

1. Appearance of cracks, record their position, and their length and depth to the nearest 1mm.
2. Number of flexing cycles.
3. Record the diameter of the smaller (i.e. flexing) roller.